

Solutions to JEE Main - 6 | JEE-2022

PHYSICS

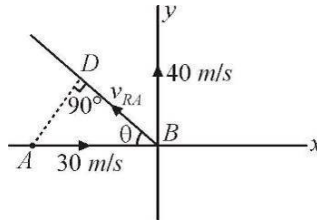
SECTION-1

1.(B) $|\overline{OC} + \overline{OA}| = \sqrt{r^2 + (\sqrt{3}r)^2} = 2r$

Direction of resultant with $\overline{OC}$, $\theta = \tan^{-1}\left(\frac{OA}{x}\right) = \tan^{-1}\left(\frac{r}{\sqrt{3}r}\right) = 30^\circ$

$\Rightarrow \overline{OA} + \overline{OC}$ lies along $\overline{OB} \Rightarrow |\overline{OA} + \overline{OB} + \overline{OC}| = 2r + r = 3r$

2.(B) $\vec{v}_{BA} = \vec{v}_B - \vec{v}_A = 40\vec{j} - 30\vec{i}$; $\tan\theta = \frac{40}{30}$



$|\vec{v}_{BA}| = 50 \text{ m/s}$

Required time = $\frac{BD}{v_{BA}} = \frac{AB \cos \theta}{v_{BA}} = \frac{6000}{50} = 120 \text{ s}$

3.(B) Since, ball gains constant velocity

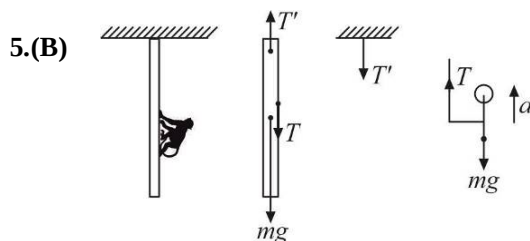
Therefore, net force on the ball = zero

$\Rightarrow$ Weight of the ball = Buoyant force + Viscous force (F_v)

$\Rightarrow Mg = \frac{M}{d_1} \times d_2 \times g + F_v \quad \therefore F_v = Mg \left(1 - \frac{d_2}{d_1}\right)$

4.(B) $X_{cm} = \frac{MX - m_1 x_1}{M - m_1} = \frac{\sigma a^2(0) - \sigma \pi b^2(c)}{\sigma \pi a^2 - \sigma \pi b^2}$

$X_{cm} = \frac{-cb^2}{(a^2 - b^2)}$



$T - mg = ma$

$T = M(g + a)$

$T' = T + mg$

$= M(g + a) + mg$

- 6.(D) In reason c to d slope dx/dt is +ve, velocity is +ve

But $\frac{d\left(\frac{dx}{dt}\right)}{dt}$ is -ve because slope decreasing so $\frac{d\left(\frac{dx}{dt}\right)}{dt} = \frac{d^2x}{dt^2}$ is -ve,

Therefore acceleration is -ve.

- 7.(D) Let σ = mass per unit area $= \frac{m}{\pi(R^2 - r^2)}$

If we fill the hole with mass densities $+\sigma$ and $-\sigma$, then the system can be treated as a combination of two discs of radii r and R and densities $-\sigma$ and $+\sigma$

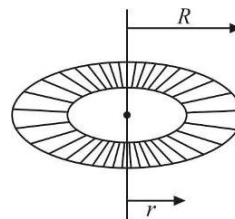
$$I = I_1 + I_2$$

$$= \frac{1}{2} m_1 R_1^2 + \frac{1}{2} m_2 R_2^2$$

$$= \frac{1}{2} (-\sigma) \pi r^2 (r^2) + \frac{1}{2} (+\sigma) \pi R^2 (R^2)$$

$$= \frac{\sigma \pi}{2} (R^4 - r^4)$$

$$= \frac{m(R^4 - r^4)}{2(R^2 - r^2)} = \frac{m}{2} (R^2 + r^2)$$



- 8.(A) $a = 2v^2 \Rightarrow \frac{dv}{dt} = 2v^2$ or $\frac{dv}{dx} \times \frac{dx}{dt} = 2v^2$

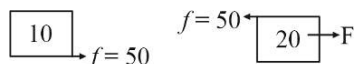
$$v \frac{dv}{dx} = 2v^2 \Rightarrow \frac{dv}{dx} = 2v$$

$$\int_{v_0}^v \frac{dv}{v} = \int_0^x 2 dx \Rightarrow [\ln v]_{v_0}^v = [2x]_0^x$$

$$\ln \frac{v}{v_0} = 2x \Rightarrow v = v_0 e^{2x}$$

- 9.(B) 10 kg will be accelerated by friction

$$f_{\max} = \mu N = 0.5 \times 10 \times 10 = 50 N$$



$$F - 50 = 20a \quad \dots\dots\dots(1)$$

$$f = 10a \quad \dots\dots\dots(2)$$

$$50 = 10a$$

$$\therefore a = 5 \text{ m/s}^2 \quad \text{(maximum acceleration for blocks to move together)}$$

$$\text{Hence } F = 50 + 20 \times 5 = 150 \text{ N}$$

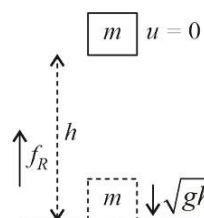
$$\therefore F_{\max} = 150 \text{ N}$$

- 10.(C) Identify initial and final state and identify all forces

$$W_g + W_{\text{air res.}} = \Delta K$$

$$mgh + W_{\text{air res.}} = \frac{1}{2} m (\sqrt{gh})^2 - 0$$

$$\Rightarrow W_{\text{air res.}} = -\frac{mgh}{2}$$



- 11.(A) The projected speed is less than escape speed $V_e = \sqrt{2gR_e}$. The body will not reach infinity.

At maximum height the kinetic energy becomes zero.

Using conservation of energy,

$$\begin{aligned} \frac{mV^2}{2} - \frac{GM_e m}{R_e} &= -\frac{GM_e m}{r} \\ \Rightarrow \frac{mgR_e}{2} - \frac{GM_e m}{R_e} &= -\frac{GM_e m}{r} \quad (GM_e = gR_e^2) \\ \Rightarrow \frac{mgR_e}{2} - mgR_e &= -\frac{gR_e^2 m}{r} \Rightarrow \frac{mgR_e}{2} = \frac{gR_e^2 m}{r} \\ \Rightarrow r &= 2R_e \quad (r = R_e + h) \quad \therefore \text{The height attained} = R_e \end{aligned}$$

- 12.(C) Extension $\propto g \left(g' \propto \frac{1}{r^2} \right)$

$$\frac{2}{x} = \frac{(3R_e)^2}{R_e^2} \Rightarrow x = \frac{2}{9} \text{ cm}$$

- 13.(B) Initially the acceleration of rocket is $2g$ upward.

After time t_0 , the speed of rocket will be

$$v = u + at \quad \text{or} \quad v = 2gt_0 \quad [u = 0]$$

$$\text{The height attained in this time} = \frac{2gt_0^2}{2} \left[\text{using } s = ut + \frac{1}{2}at^2 \right]$$

After time t_0 , the engine is switched off. Now it will be a case of retardation g . The initial velocity now will be $2gt_0$.

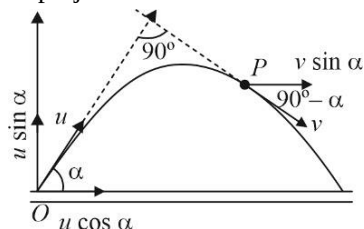
$$\text{The height at which it will stop will be } \frac{(2gt_0)^2}{2g} \left[\text{using } H_{\max} = \frac{u^2}{2g} \right]$$

$$\text{Total height attained} = gt_0^2 + 2gt_0^2 = 3gt_0^2$$

- 14.(B) Horizontal velocity at point 'O' = $u \cos \alpha$

Horizontal velocity at point 'P' = $v \sin \alpha$

In projectile motion horizontal component of velocity remains constant throughout the motion



$$\therefore v \sin \alpha = u \cos \alpha \Rightarrow v = u \cot \alpha$$

- 15.(C) Tangential acceleration,

$$a_t = \frac{dv}{dt} = \frac{d}{dt}(4t) = 4 \text{ cm s}^{-2}$$

$$a_c = \frac{v^2}{R} = \frac{(4)^2}{2} = 8 \text{ cm s}^{-2}$$

$$\therefore a = \sqrt{a_t^2} = \sqrt{(4)^2 + (8)^2} = 4\sqrt{5} \text{ cm s}^{-2}$$

- 16.(C) Let v be the linear velocity of centre of mass of C and ω its angular velocity about centre of mass.

$$\text{Then, } \omega = \frac{v}{2R}$$

$$\begin{aligned} \text{K.E. of spherical body, } K_1 &= \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 \\ &= \frac{1}{2}mv^2 + \frac{1}{2}(2mR^2)\left(\frac{v^2}{4R^2}\right) = \frac{3}{4}mv^2 \end{aligned} \quad \dots\dots\dots(i)$$

Speed of the block will be

$$v' = (\omega)(3R) = 3R\omega = (3R)\left(\frac{v}{2R}\right) = \frac{3}{2}v$$

$$\therefore \text{ K. E. of block } K_2 = \frac{1}{2}mv'^2 = \frac{1}{2}m\left(\frac{3}{2}v\right)^2 = \frac{9}{8}mv^2 \quad \dots\dots\dots(ii)$$

$$\text{From equations (i) and (ii), } \frac{K_1}{K_2} = \frac{2}{3}$$

- 17.(A) The height of water in the tank becomes maximum when the volume of water flowing into the tank per second becomes equal to the volume flowing out per second.

$$\text{Volume of water flowing out per second} = Av = A\sqrt{2gh} \quad \dots\dots\dots(i)$$

$$\text{Volume of water flowing in per second} = 70 \text{ cm}^3/\text{sec} \quad \dots\dots\dots(ii)$$

From (i) and (ii), we get

$$A\sqrt{2gh} = 70 \Rightarrow 1 \times \sqrt{2gh} = 70 \Rightarrow 1 \times \sqrt{2 \times 980 \times h} = 70 \quad \therefore h = \frac{4900}{1960} = 2.5 \text{ cm}$$

- 18.(B) For the given situation, liquid of density 2ρ should be behind that ρ from right limb:

$$P_A = P_{atm} + \rho gh$$

$$P_B = P_A + \rho a \frac{l}{2} = P_{atm} + \rho gh + \rho a \frac{l}{2}$$

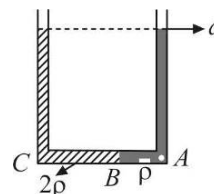
$$P_C = P_B + (2\rho)a \frac{l}{2} = P_{atm} + \rho gh + \frac{3}{2}\rho al \quad \dots(1)$$

But from left limb:

$$P_C = P_{atm} + (2\rho)gh \quad \dots(2)$$

From (1) and (2)

$$P_{atm} + \rho gh + \frac{3}{2}\rho al = P_{atm} + 2\rho gh \Rightarrow h = \frac{3a}{2g}l$$



$$19.(B) P_B - P_0 = \frac{2T}{3R}, P_A = P_B + \frac{2T}{R}, P_A - P_0 = \frac{2T}{3R} + \frac{2T}{R} = 2T\left[\frac{4}{3R}\right], \frac{P_B - P_0}{P_A - P_0} = \frac{1}{4}$$

- 20.(B) FBD of C $kx_1 = mg$

$$\text{FBD of } B, T = 2mg$$

$$\text{FBD of } A, kx_2 = mg + T$$

$$kx_2 = mg$$

When string is cut out $T = 0$

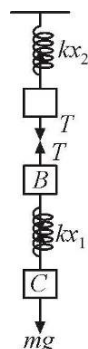
$$kx_2 - mg = ma_A$$

$$3mg - mg = ma_A \Rightarrow a_A = 2g$$

For B

$$T - mg - kx_1 = ma_B$$

$$0 - 2mg = ma_B ; a_B = -2g$$



SECTION-2

21.(2) Resultant momentum $= \sqrt{(2)^2 + (2)^2} = 2\sqrt{2}$

$$\text{Mass} = \frac{\text{Momentum}}{\text{Velocity}} = \frac{2\sqrt{2}}{\sqrt{2}} = 2 \text{ kg}$$

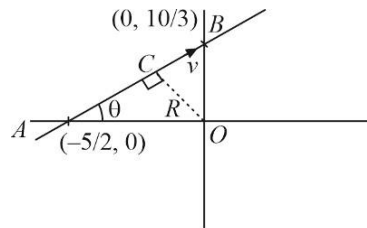
22.(8) Angular momentum of particle moving in line AB is $L = mvR$

$$\tan \theta = \frac{10 \times 2}{3 \times 5} = \frac{4}{3}$$

$$\theta = 53^\circ$$

$$\text{In } \triangle OAC, \sin 53^\circ = \frac{R}{5/2}$$

$$\Rightarrow R = \frac{5}{2} \times \frac{4}{5} = 2 \Rightarrow L = 1 \times 4 \times 2 = 8 \text{ Kg m}^2/\text{s}$$



23.(5) Velocity of the centre $= v_0$ and the angular velocity about the centre $= \frac{v_0}{2R}$. Thus $v_0 > \omega_0 R$. The

sphere slips forward and thus the friction by the plane on the sphere will act backward. As the friction is kinetic, its value is $\mu N = \mu Mg$ and the sphere will be decelerated by $a_{cm} = f/M$.

$$\text{Hence, } v(t) = v_0 - \frac{f}{M}t \dots\dots\dots(i)$$

This friction will also have a torque $\tau = fr$ about the centre. This torque is clockwise and in the direction of ω_0 . Hence the angular acceleration about the centre will be

$$\alpha = f \frac{R}{(2/3)MR^2} = \frac{3f}{2MR}$$

$$\text{And the clockwise angular velocity at time } t \text{ will be } \omega(t) = \omega_0 + \frac{3f}{2MR}t = \frac{v_0}{2R} + \frac{3f}{2MR}t$$

$$\text{Pure rolling starts when } v(t) = R\omega(t) \text{ i.e., } v(t) = \frac{v_0}{2} + \frac{3f}{2M}t \dots\dots\dots(ii)$$

$$\text{Eliminating } t \text{ from (i) and (ii), } \frac{3}{2}v(t) + v(t) = \frac{3}{2}v_0 + \frac{v_0}{2} \text{ or } v(t) = \frac{2}{5} \times 2v_0 = \frac{4}{5}v_0$$

Thus, the sphere rolls with linear velocity $4v_0/5$ in the forward direction.

24.(9.60) Here, velocity at the top most point is zero.

From energy conservation

$$\text{Let } v_B = v; \omega = \frac{v}{2l}; v_A = \omega l = \frac{v}{2}$$

$$\frac{1}{2}mv^2 + \frac{1}{2}m\frac{v^2}{4} = mg(2l) + mg(4l) + 0 \Rightarrow v = \sqrt{\frac{48gl}{5}} \Rightarrow x = 9.6$$

25.(60) Let the speed of the bullet be v . Let the common velocity of the bullet and the bob, after the bullet is embedded into the bob, be V . By the principle of conservation of linear momentum,

$$V = \frac{(0.05 \text{ kg})v}{0.45 \text{ kg} + 0.05 \text{ kg}} = \frac{v}{10}$$

The string becomes loose and the bob will go up with a deceleration of $g = 10 \text{ m/s}^2$. As it comes to rest at a height of 1.8 m, using the equation $v^2 = u^2 + 2ax$, we get

$$1.8 \text{ m} = \frac{\left(\frac{v}{10}\right)^2}{2 \times 10 \text{ m/s}^2} \text{ or } v = 60 \text{ m/s}$$

Chemistry

SECTION-1

1.(C) Moles of $\text{KNO}_3 = \text{moles of } \text{NO}_3^- = \frac{10.1}{101} = 0.1$

$\therefore$ Density of $\text{H}_2\text{O} = 1 \text{ g mL}^{-1}$

Weight of $\text{H}_2\text{O} = 500 \text{ mL} \times 1 = 500 \text{ g} = 0.5 \text{ kg}$

$$m_{\text{NO}_3^-} = \frac{\text{Moles of } \text{NO}_3^-}{\text{Weight of solvent } (\text{H}_2\text{O}) \text{ in kg}} = \frac{n_{\text{NO}_3^-}}{0.5 \text{ kg}}$$

$$\Rightarrow 0.3 = \frac{n_{\text{NO}_3^-}}{0.5 \text{ kg}} \quad \therefore n_{\text{NO}_3^-} = 0.3 \times 0.5 = 0.15$$

(required)

Moles of NO_3^- obtained from $\text{KNO}_3 = 0.1$

Moles of NO_3^- required from $\text{Ba}(\text{NO}_3)_2 = 0.15 - 0.1 = 0.05$

[2 mol of NO_3^- is obtained from 1 mol of $\text{Ba}(\text{NO}_3)_2$]

$\therefore$ Moles of $\text{Ba}(\text{NO}_3)_2 = \frac{0.05}{2}$

Weight of $\text{Ba}(\text{NO}_3)_2 = \frac{0.05}{2} \times 261 = 6.5 \text{ g}$

2.(C) (i) 1 molecule of $\text{O}_2 = \frac{32}{6 \times 10^{23}}$

(ii) 1 atom of N = $\frac{14}{6 \times 10^{23}}$

(iii) 1 mol of $\text{H}_2\text{O} = 18 \text{ g}$

(iv) Weight of Fe = 10^{-10} g

So, the order is (C).

3.(B) $\lambda \propto \frac{1}{\sqrt{\text{KE}}} = \sqrt{\frac{\text{KE}_2}{\text{KE}_1}} = \frac{\lambda_1}{\lambda_2}$ or $\frac{\lambda_1}{0.99\lambda_1}$

$$\Rightarrow \text{KE}_2 = \frac{\text{KE}_1}{(0.99)^2} = 1.02 \text{ KE}_1$$

% change in KE = $\frac{\text{KE}_2 - \text{KE}_1}{\text{KE}_1} \times 100 \approx 2\%$

4.(A) Frequency $\nu = \frac{V}{2\pi r}$; $V \propto \frac{Z}{n}$ and $r \propto \frac{n^2}{Z}$

$$\nu = \frac{Z \times Z}{n \times n^2} \Rightarrow \nu \propto \frac{Z^2}{n^3}$$

So $\frac{\nu_{\text{He}^+}}{\nu_{\text{H}}} = \frac{2^2/3^3}{1^2/2^3} = \frac{8 \times 4}{27} = \frac{32}{27}$

Second excited state $\Rightarrow n = 3$

5.(D) The velocity of X-rays is same as that of microwaves.

6.(D) CO gas, because root mean square velocity $= \left(\frac{3RT}{M} \right)^{1/2}$

i.e., $\text{RMS} \propto \left(\frac{1}{M} \right)^{1/2}$

CO has the lowest molecular mass; hence, it has the highest RMS velocity.

7.(D) (A) Density of gas $(\rho) = \frac{PM}{RT}$

Since $\frac{M_B}{T_B} = \frac{M_A}{T_A}$,

Therefore, at the same pressure $\rho_A = \rho_B$. But if pressure is different, then $\rho_A \neq \rho_B$.

(B) The pressure of the gases will be equal if their densities are equal, otherwise not [as in case (A)]

(C) KE per mole $= \frac{3}{2} RT$

Hence, it will be different for given gases.

(D) $u_{\text{rms}} = \sqrt{\frac{3RT}{M_w}}$

Since $\frac{T_A}{M_A} = \frac{T_B}{M_B}$, u_{rms} of A $= u_{\text{rms}}$ of B

8.(D) Ideal gas cannot be liquefied at any value of P and T.

9.(D) $b = 4 \times \text{volume occupied by the molecules in 1 mol of gas}$

$$= 4 \times \frac{4}{3} \times \pi r^3 \times N_A = 4 \times \frac{4}{3} \times 3.14 \times (1 \times 10^{-8})^3 \times 6 \times 10^{23} = 100.4 \times 10^{-1} \approx 10 \text{ mL mol}^{-1}$$

10.(C) $VT^3 = \text{constant}$

Also $V^{\gamma-1} T = \text{constant}$ for adiabatic expansion

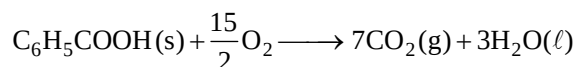
$$V \cdot T^{\frac{1}{\gamma-1}} = \text{constant} \text{ or } \frac{1}{\gamma-1} = 3$$

$$\gamma = \frac{4}{3}, \text{ i.e., polyatomic gas}$$

11.(B) In the process of formation, the compound must be formed from constituent elements in their standard state.

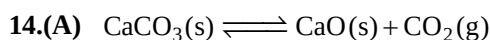
12.(A) $\Delta S_{\text{total}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0$

13.(C) $\Delta H^\ominus = \Delta U^\ominus + \Delta n_g RT$



$$\Delta n_g = n_p - n_R = 7 - \frac{15}{2} = \frac{-1}{2}$$

$$\therefore \Delta H^\ominus = -321.30 + \left[\frac{-1}{2} \times 300 \text{ K} \times R \right] = -321.30 - 150 R$$



$$K_p = P_{\text{CO}_2}$$

$$\Delta G^\circ = -2.303 RT \log K_p$$

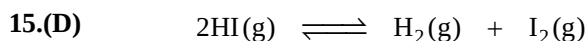
$$\Delta H^\circ - T\Delta S^\circ = -2.303 RT \log K_p$$

$$\log K_p = -\frac{\Delta H^\circ}{2.303RT} + \frac{\Delta S^\circ}{2.303R}$$

$$\log P_{\text{CO}_2} = -\frac{\Delta H^\circ}{2.303RT} + \frac{\Delta S^\circ}{2.303R} \quad \dots\dots\dots(i)$$

$$y = mx + c$$

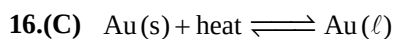
Graph (A) represents (i) and its slope will be used to determine the heat of the reaction.



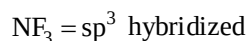
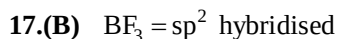
At eq. $1 - \alpha \qquad \frac{\alpha}{2} \qquad \frac{\alpha}{2}$

$$K_p = \frac{\left(\frac{\alpha}{2} P_T\right)^2}{(1 - \alpha)^2 P_T^2} \qquad \left(\because K_p = \frac{P_{\text{I}_2} \times P_{\text{H}_2}}{(P_{\text{HI}})^2} \right)$$

$$\frac{\alpha}{1 - \alpha} = 2\sqrt{K_p} \Rightarrow \alpha = \frac{2\sqrt{K_p}}{1 + 2\sqrt{K_p}}$$

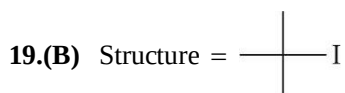


There is an increase in volume in the forward direction, hence the melting is favoured at low pressure.



% s character is more in hybrid orbital of BF_3 than NH_3

18.(A) All given species have 14 electrons.



20.(C) Relation between ΔH (enthalpy change) and ΔE (internal energy change) is

$$\Delta H = \Delta E + \Delta n_g RT$$

Where, $\Delta n_g = (\text{moles of gaseous products}) - (\text{moles of gaseous reactants})$

For the given reaction, $\Delta n_g = 2 - 3 = -1$

$$\Rightarrow -1366.5 = \Delta E - 1 \times 8.314 \times 10^{-3} \times 300$$

$$\therefore \Delta E = -1364.0 \text{ kJ mol}^{-1}$$

SECTION-2

21.(7.20) 1 mole of $\text{Na}_3[\text{Fe}(\text{CN})_6]$ contains = 6 mol of C

0.1 mole of $\text{Na}_3[\text{Fe}(\text{CN})_6]$ contains = 0.6 mol of C

$$= 0.6 \times 12 = 7.2 \text{ g}$$

22.(12) If each orbital can accommodate 3 electron, then atomic number of Ne will be 19

Its electronic configuration will be $1s^3 2s^3 2p^9$

$\therefore$ Number of electrons having $n = 2$ are 12

23.(0.34) Let the molecular mass of the mixture at equilibrium be M_{mix} .

Applying the relation :

$$M_{\text{mix}} = \frac{dRT}{P} = \frac{0.925 \times 0.0821 \times 900}{1} = 68.348$$

Molecular mass of $\text{SO}_3 = 80$

$$\text{Vapour density of } \text{SO}_3, D = \frac{80}{2} = 40$$

$$\text{Vapour density of mixture, } d = \frac{68.348}{2} = 34.174$$

Let the degree of dissociation be x .

$$x = \frac{D - d}{(n - 1)d} = \frac{40 - 34.174}{\left(\frac{3}{2} - 1\right) \times 34.174} = \frac{5.826 \times 2}{34.174} = 0.34$$

24.(0) From MO diagram.

25.(5) Except ClF_3 , all are linear non-polar molecules.

Mathematics

SECTION-1

- 1.(C) Let the mid-point of chords of circle be P(h, k).

Equation of this chord is $T = S_1$

$$\text{i.e., } xh + yk + g(x+h) + f(y+k) + c = h^2 + k^2 + 2gh + 2fk + c$$

It passes through (0, 0) thus $h^2 + k^2 + gh + fk = 0$

i.e., required locus is $x^2 + y^2 + gx + fy = 0$

- 2.(A) Let $P \equiv (a, 25-a)$. Equation of chord AB is $T = 0$ i.e., $xa + y(25-a) = 9$. If mid-point of chord AB

is C(h, k), then equation of chord AB is $T = S_1$ i.e., $xh + yk = h^2 + k^2$. Comparing the coefficient of like powers, we get

$$\frac{a}{h} = \frac{25-a}{k} = \frac{9}{h^2 + k^2} = \frac{a + 25 - a}{h + k} = \frac{25}{h + k} \Rightarrow 9(h + k) = 25(h^2 + k^2) \Rightarrow 25(x^2 + y^2) = 9(x + y)$$

$$\begin{aligned} 3.(D) \quad S &= -i \sum_{k=1}^6 \left(\cos \frac{2k\pi}{7} + i \sin \frac{2k\pi}{7} \right) \\ S &= -i \sum_{k=0}^6 \left(\cos \frac{2k\pi}{7} + i \sin \frac{2k\pi}{7} \right) + i \\ &= 0 + i = i \end{aligned}$$

- 4.(A) As $1, z \neq 1, z = \cos \theta + i \sin \theta, -\pi < \theta \leq \pi, \theta \neq 0$, now

$$\begin{aligned} \omega &= \frac{1}{1-z} = \frac{1}{1-\cos \theta + i \sin \theta} = \frac{(1-\cos \theta) + i \sin \theta}{(1-\cos \theta)^2 + \sin^2 \theta} = \frac{(1-\cos \theta) + i \sin \theta}{2(1-\cos \theta)} \\ &= \frac{1}{2} + \frac{i}{2} \cot \left(\frac{\theta}{2} \right) = \frac{1}{2} + \frac{i}{2} \tan \left(\frac{\pi}{2} - \frac{\theta}{2} \right) \end{aligned}$$

This shows that ω lies on the line $x = 1/2$ and $-\frac{\pi}{2} < \arg(\omega) \leq \frac{\pi}{2}$, $\arg(\omega) \neq 0$, the maximum value of $\arg(\omega)$ is never attained.

- 5.(D) We have $\alpha + \beta = -p, \alpha\beta = q$

$$\text{Now, } (\omega\alpha + \omega^2\beta)(\omega^2\alpha + \omega\beta) = \omega^3\alpha^2 + \omega^4\alpha\beta + \omega^2\alpha\beta + \omega^3\beta^2$$

$$\alpha^2 + \beta^2 + (\omega + \omega^2)\alpha\beta = \alpha^2 + \beta^2 - \alpha\beta = (\alpha + \beta)^2 - 3\alpha\beta = p^2 - 3q$$

- 6.(A) $|z_1| = |z_2| = |z_3| = 1$

$$z_1 \bar{z}_1 = z_2 \bar{z}_2 = z_3 \bar{z}_3 = 1$$

$$\therefore |z_1 + z_2 + z_3| = \left| \frac{1}{\bar{z}_1} + \frac{1}{\bar{z}_2} + \frac{1}{\bar{z}_3} \right| = \left| \frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} \right| = 1$$

- 7.(C) Let $P \equiv (a, 4-2a)$. Equation of chord of contact is

$$x.a + y.(4-2a) = 1$$

$$(4y-1) + a(x-2y) = 0$$

It will always pass through a fixed point whose coordinates are $y = \frac{1}{4}$ and $x = 2y = \frac{1}{2}$.

8.(D) The given expression is $\frac{(e^{-i2\theta})^7 (e^{i3\theta})^{-5}}{(e^{i4\theta})^{12} (e^{i5\theta})^{-6}} = \frac{e^{-i14\theta} \cdot e^{-i15\theta}}{e^{i48\theta} e^{-i30\theta}} = \frac{e^{-i29\theta}}{e^{i18\theta}} = e^{-i47\theta} = \cos 47\theta - i \sin 47\theta$

9.(C) $z\bar{z} = |z|^2 \Rightarrow$ If z is of unit modulus, then $z\bar{z} = 1$ or $\bar{z} = \frac{1}{z}$

So, $\arg\left(\frac{1+z}{1+\bar{z}}\right)$ becomes $\arg\left(\frac{1+z}{1+1/z}\right) = \arg(z) = \theta$

10.(A) The equation of normal to the parabola $y^2 = 12x$ with slope -1 is

$$y = -x - 2(3)(-1) - 3(-1)^3 \Rightarrow y = -x + 9 \Rightarrow x + y = 9$$

$\therefore k = 9$

Since, the focus of the parabola is $(3, 0)$

$\therefore p = \left| \frac{3-9}{\sqrt{2}} \right| \Rightarrow 2p^2 = 36 \quad \therefore 3k^3 + 2p^2 = 3(9)^3 + 36 = 2223$

11.(A) Centre of $(\alpha, 2)$ and radius $= 2$

$$\frac{\left| \frac{\alpha}{2} - 2 \right|}{\sqrt{\frac{5}{2}}} = 2 \Rightarrow \left(\frac{\alpha}{2} - 2 \right) = \sqrt{5}$$

$$\frac{\alpha}{2} - 2 = \sqrt{5} \Rightarrow \alpha = 4 + 2\sqrt{5}$$

12.(D) Vertex $= (1, 1)$, Focus $= (2, 2)$. So, the directrix is $x + y = 0$.

Then, equation is $\frac{x+y}{\sqrt{2}} = \sqrt{(x-2)^2 + (y-2)^2} \Rightarrow (x-y)^2 = 8(x+y-2)$

13.(A) Since, end of latus rectum is $\left(2e, \frac{3}{2} \right)$

Normal at $\left(2e, \frac{3}{2} \right)$ is $\frac{x-2e}{2e/4} = \frac{\left(y - \frac{3}{2} \right)}{\frac{3}{2 \times 3}}$

$$\Rightarrow \frac{2(x-2e)}{e} = 2(y) - 4(1-e^2) \quad \left[\because b^2 = a^2(1-e^2) \Rightarrow 3 = 4(1-e^2) \right]$$

$$\Rightarrow 2x - 4e = 2ey + 4e^3 - 4e \Rightarrow 2x - 2ey = 4e^3$$

$$\Rightarrow (x - ey) = 2e^3 \Rightarrow x - ey - 2e^3 = 0$$

14.(C) Tangents at t and t' meet on the point (x, y) given by

$$x = att' = ak^2 t'^2 \quad \dots \text{(i)}$$

and $y = a(t + t') = a(k^2 t' + t')$

$$= at'(k^2 + 1) \quad \dots \text{(ii)}$$

From equations (i) and (ii),

$$x = \frac{ak^2 \cdot y^2}{a^2(k^2+1)^2} = \frac{k^2 y^2}{a(k^2+1)^2} \Rightarrow y^2 = \frac{ax(k^2+1)^2}{k^2} = ax \left(k + \frac{1}{k}\right)^2$$

15.(C) Given equation is $\frac{x^2}{16} + \frac{y^2}{(16/\sqrt{11})^2} = 1$

Thus, the parametric coordinates are $\left(4\cos\phi, \frac{16}{\sqrt{11}}\sin\phi\right)$

The equation of tangent at this point is $\frac{x\cos\phi}{4} + \frac{\sqrt{11}y\sin\phi}{16} = 1$

This touches the circle $x^2 + y^2 - 2x - 15 = 0$

$$\begin{aligned} \therefore \frac{\left|\frac{\cos\phi}{4} - 1\right|}{\sqrt{\frac{\cos^2\phi}{16} + \frac{11\sin^2\phi}{256}}} &= 4 \\ \Rightarrow \cos^2\phi + 16 - 8\cos\phi &= 256\left(\frac{\cos^2\phi}{16} + \frac{11\sin^2\phi}{256}\right) \Rightarrow \\ 15\cos^2\phi + 11(1 - \cos^2\phi) + 8\cos\phi - 16 &= 0 \\ \Rightarrow 4\cos^2\phi + 8\cos\phi - 5 &= 0 \\ \Rightarrow \cos\phi = \frac{1}{2} \quad \left[\because \cos\phi \neq \frac{-5}{2}\right] &\Rightarrow \phi = \frac{\pi}{3} \quad \text{or} \quad -\frac{\pi}{3} \end{aligned}$$

16.(A) Parabola intersect y-axis at (0, 2), (0, 3)

17.(A) Equation of chord of contact 'AB' $\equiv hx + ky = 1$

AB is common chord also

$$\Rightarrow (\lambda + 6)x - (8 - 21)y + 2 = 0$$

$$\frac{\lambda + 6}{h} = \frac{2\lambda - 8}{k} = \frac{2}{-1} \Rightarrow \frac{(2\lambda + 12) - (2\lambda - 8)}{2h - k}$$

$$2h - k = -10$$

$$2x - y + 10 = 0$$

18.(D) Let the equations of circle be $S_1 \equiv x^2 + y^2 - 4x - 4y + 4 = 0$ and $S_2 \equiv x^2 + y^2 - x - y + \frac{1}{4} = 0$

Equation of common chord AB is $S_1 - S_2 = 0$

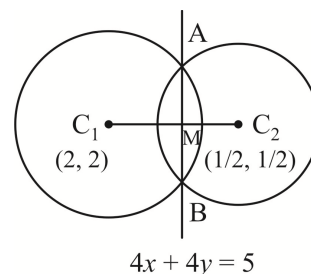
$$\Rightarrow 4x + 4y = 5 \quad \dots (i)$$

Now, $C_1M = \frac{4(2) + 4(2) - 5}{\sqrt{4^2 + 4^2}} = \frac{11}{4\sqrt{2}}$ and $r_1 = \sqrt{4 + 4 - 4} = 2$

Also equation of line C_1C_2 is $y = x \quad \dots (ii)$

On solving equation (i) and (ii), we get : $x = \frac{5}{8}, y = \frac{5}{8}$

Radius of required circle, $AM = \sqrt{AC_1^2 - C_1M^2} = \sqrt{4 - \frac{121}{32}} = \frac{\sqrt{7}}{4\sqrt{2}}$



19.(A) $z = -2 + 2\sqrt{3}i = 4\omega$

$$\begin{aligned} z^{2n} + 2^{2n} \cdot z^n + 2^{4n} &= 4^{2n} \cdot \omega^{2n} + 2^{2n} \cdot 4^n \cdot \omega^n + 2^{4n} \\ &= 4^{2n} [\omega^{2n} + \omega^n + 1] \\ &= \begin{cases} 0, & \text{if } n \text{ is not multiple of } 3 \\ 3 \cdot 4^{2n}, & \text{if } n \text{ is multiple of } 3 \end{cases} \end{aligned}$$

20.(C) Given that, $z_r = \cos \frac{r\alpha}{n^2} + i \sin \frac{r\alpha}{n^2}$ [where, $r = 1, 2, 3, \dots, n$]

$$z_1 = \cos \frac{\alpha}{n^2} + i \sin \frac{\alpha}{n^2}$$

$$z_2 = \cos \frac{2\alpha}{n^2} + i \sin \frac{2\alpha}{n^2}$$

$$\vdots$$

$$z_n = \cos \frac{n\alpha}{n^2} + i \sin \frac{n\alpha}{n^2}$$

$$\begin{aligned} \therefore \lim_{n \rightarrow \infty} (z_1 z_2 \dots z_n) &= \lim_{n \rightarrow \infty} \left(\cos \frac{\alpha}{n^2} + i \sin \frac{\alpha}{n^2} \right) \times \left(\cos \frac{2\alpha}{n^2} + i \sin \frac{2\alpha}{n^2} \right) \dots \left(\cos \frac{n\alpha}{n^2} + i \sin \frac{n\alpha}{n^2} \right) \\ &= \lim_{n \rightarrow \infty} \left[\cos \left\{ \frac{\alpha}{n^2} (1 + 2 + 3 + \dots + n) \right\} + i \sin \left\{ \frac{\alpha}{n^2} (1 + 2 + 3 + \dots + n) \right\} \right] \\ &= \lim_{n \rightarrow \infty} \left[\cos \frac{\alpha n(n+1)}{2n^2} + i \sin \frac{\alpha n(n+1)}{2n^2} \right] = \cos \frac{\alpha}{2} + i \sin \frac{\alpha}{2} = e^{\frac{i\alpha}{2}} \end{aligned}$$

SECTION-2

21.(3) We have $|x|^2 = x\bar{x} = (a+b+c)(\bar{a}+\bar{b}+\bar{c}) = |a|^2 + |b|^2 + |c|^2 + \bar{a}(b+c) + \bar{b}(c+a) + \bar{c}(a+b) \dots\dots(i)$

Similarly $|y|^2 = |a|^2 + |b\omega|^2 + \bar{a}(b\omega + c\omega^2) + \bar{b}\omega(c\omega^2 + a) + \bar{c}\bar{\omega}(a + b\omega) \dots\dots(ii)$

$$= |a|^2 + |b|^2 + |c|^2 + \bar{a}(b\omega + c\omega^2) + \bar{b}(c\omega + a\omega^2) + \bar{c}(a\omega + b\omega^2)$$

And $|z|^2 = |a|^2 + |b|^2 + |c|^2 + \bar{a}(b\omega^2 + c\omega) + \bar{b}(a\omega + c\omega^2) + \bar{c}(a\omega^2 + b\omega) \dots\dots(iii)$

Adding (i), (ii) and (iii), we obtain

$$|x|^2 + |y|^2 + |z|^2 = 3(|a|^2 + |b|^2 + |c|^2)$$

$$\Rightarrow \frac{|x|^2 + |y|^2 + |z|^2}{|a|^2 + |b|^2 + |c|^2} = 3$$

22.(12) $z^2 + z + 1 = 0 \Rightarrow z = \omega, \omega^2$.

Let $z = \omega$ then

$$(\omega + \omega^2)^2 = (\omega^2 + \omega^4)^2 = (\omega^4 + \omega^8)^2 = (\omega^5 + \omega^{10})^2 = 1 \text{ and } (\omega^3 + \omega^6)^2 = (\omega^6 + \omega^{12})^2 = 4$$

Thus, $S = 12$.

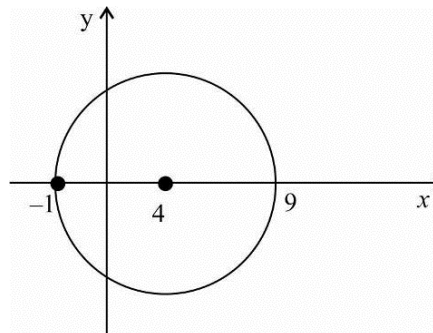
$$23.(3) \quad \frac{1}{x+iy} = \frac{\cos\theta + 2}{3} + i \frac{\sin\theta}{3}$$

$$\Rightarrow \quad \frac{x-iy}{x^2+y^2} = \frac{\cos\theta + 2}{3} + i \frac{\sin\theta}{3}$$

$$\Rightarrow \quad \left(\frac{x}{x^2+y^2} - \frac{2}{3} \right)^2 + \left(\frac{-y}{x^2+y^2} \right)^2 = \frac{\cos^2\theta + \sin^2\theta}{9}$$

$$\Rightarrow \quad \frac{x^2+y^2}{(x^2+y^2)^2} - \frac{4x}{3(x^2+y^2)} + \frac{4}{9} = \frac{1}{9}$$

$$\Rightarrow \quad 3 - 4x + x^2 + y^2 = 0$$



$$24.(1) \quad C_1C_2 = |r_2 - r_1|$$

$\Rightarrow$ Circles touch internally

Hence, the number of common tangents is one.

25.(8.08) Tangent to $\frac{x^2}{25} + \frac{y^2}{4} = 1$ is $\frac{x}{5}\cos\theta + \frac{y}{2}\sin\theta = 1$. If it is also tangent to the circle, then

$$16 = \frac{1}{\frac{\cos^2\theta}{25} + \frac{\sin^2\theta}{4}} = \frac{100}{4 + 21\sin^2\theta} \quad \Rightarrow \quad \sin^2\theta = \frac{3}{28}, \cos^2\theta = \frac{25}{28}$$

If the tangents meets the axes at A and B, then $A = \left(\frac{5}{\cos\theta}, 0 \right)$ and $B = \left(0, \frac{2}{\sin\theta} \right)$

$$\therefore \quad AB^2 = \frac{25}{\cos^2\theta} + \frac{4}{\sin^2\theta} = 25 + \frac{4}{3} \cdot 28 = \frac{196}{3} \quad \Rightarrow \quad AB = \frac{14}{\sqrt{3}}$$